Result 1 (graded chain component). [Topology foundations, graded chain homology component]
Let K be a type. Let *field* be a field structure on K. Let C and D be elements of Ch_K. Let f be a chain map from C to D. Let p be a natural number. Then

$$H_*(f).\mathrm{component}(p) = H_p(f).$$

Result 2 (graded chain identity). [Topology foundations, homology functor identity]
Let K be a type. Let *field* be a field structure on K. Let C be a chain complex of modules over K in natural degrees. Then

$$H_*(\mathrm{id}_C) = \mathrm{id}_{H_*(C)}\,.$$

Adapted from the cited source.

Result 3 (graded chain composition). [Topology foundations, homology functor composition]
Let K be a type. Let *field* be a field structure on K. Let C, D and E be elements of Ch_K. Let f be a chain map from C to D. Let g be a chain map from D to E. Then

$$H_*(g \circ f) = (H_*(g) \circ H_*(f)).$$

Adapted from the cited source.

Result 4 (graded relative component). [Topology foundations, relative homology in the selected degree]
Let K be a type. Let *field* be a field structure on K. Let H be a choice of coefficients and homology degree over K. Let X be a topological space. Let P be two nested subsets of X with inclusion evidence. Then

$$H_*(P; H).\mathrm{component}(H.\mathrm{degree}) = H(P).$$

Result 5 (graded relative map component). [Topology foundations, relative homology map in the selected degree]
Let K be a type. Let *field* be a field structure on K. Let H be a choice of coefficients and homology degree over K. Let X be a topological space. Let Q be a certified inclusion between pairs in X. Assume the chain square of Q commutes (`h`). Then

$$H_*(Q; H).\mathrm{component}(H.\mathrm{degree}) = H(Q).$$

LLM-assisted example and proof (gpt-6, 2026-11-06). Mathematical sources are attributed above; see PROVENANCE.md.